

Z-Scores: A Metric for Linguistically Assessing Disfluency Removal

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Disfluencies



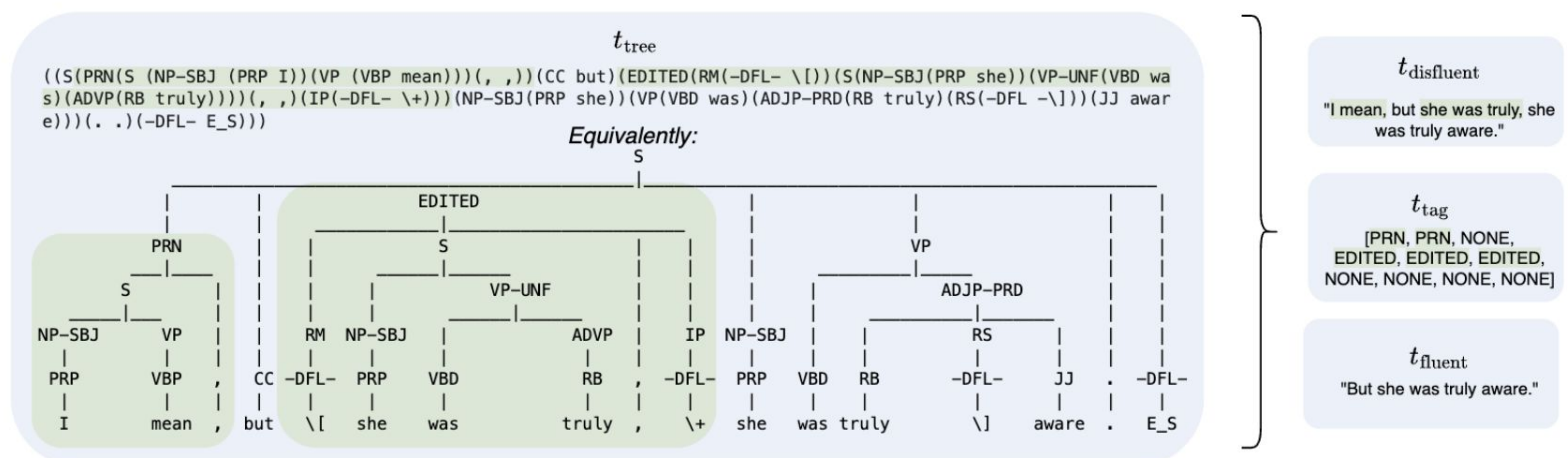
Results

		gpt-4o-mini					
M		$\mathcal{E}_F$	$\mathcal{E}_P$	$\mathcal{E}_R$	$\mathcal{Z}_E$	$\mathcal{Z}_I$	$\mathcal{Z}_P$
s	P_0	72.69 _{5.79}	75.61 _{7.05}	70.48 _{7.35}	85.20 _{8.23}	61.89 _{11.08}	65.02 _{20.99}
s	P_1	81.94 _{3.75}	84.47 _{4.92}	79.90 _{5.65}	83.67 _{9.27}	78.28 _{8.10}	74.86 _{22.06}
s	P_2	79.86 _{5.42}	76.88 _{7.02}	83.52 _{6.12}	87.45 _{7.48}	79.60 _{8.89}	87.09 _{15.46}

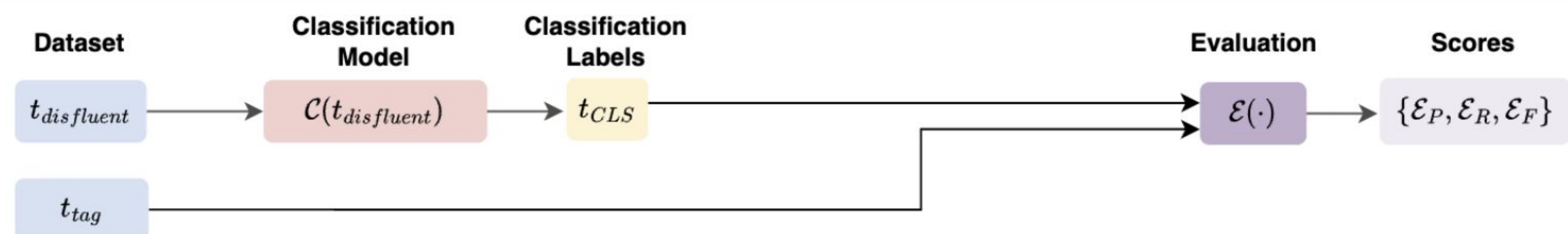
Metaprompting Experiment. We provide examples of the most-failed-on categories of disfluencies, and see that performance improves.

Z-Score Framework

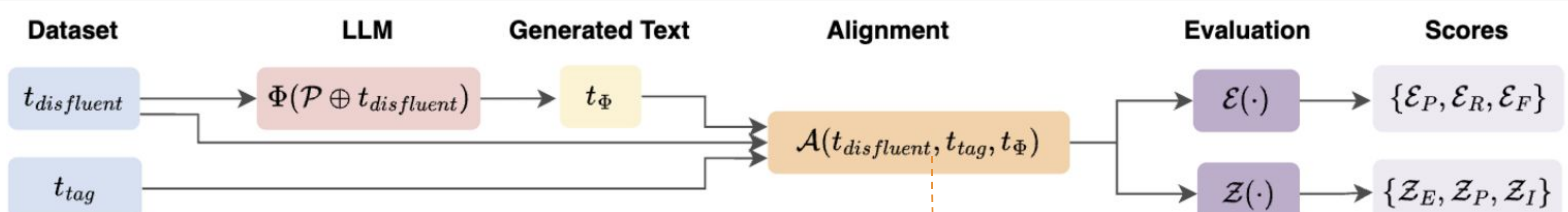
Data Pre-Processing: $T = \left\{ \left(t_{tree}^{(i)}, t_{fluent}^{(i)}, t_{tag}^{(i)}, t_{disfluent}^{(i)} \right) \right\}_{i=1}^N$ from $t_{tree}^{(i)}$ using disfluent node types: $\{EDITED, PRN, INTJ\}$.



Previous Work: Classify each word in the original input as fluent (O) or disfluent (I) to evaluate disfluency removal performance in terms of $\mathcal{E}$ -Scores.



Our Method: Align LLM output with original input to evaluate disfluency removal performance in terms of $\mathcal{E}$ -Scores & $\mathcal{Z}$ -Scores.



Alignment Module, enables scoring of generative models, and complements previous E-Score metrics.

$t_{disfluent}$	t_{tag}	t_{Φ}	t_{CLS}	$\mathbb{1}_{gt}$	$\mathbb{1}_{pred}$	$\mathbb{1}_{tp}$	$\mathbb{1}_{tn}$	$\mathbb{1}_{fp}$	$\mathbb{1}_{fn}$
i	PRN	i	I	1	0	0	0	0	1
mean	PRN	mean	I	1	0	0	0	0	1
but	NONE	but	O	0	0	0	1	0	0
she	EDITED		I	1	1	1	0	0	0
was	EDITED		I	1	1	1	0	0	0
truly	EDITED		I	1	1	1	0	0	0
she	NONE		I	0	1	0	0	1	0
-	-	Luna	O	*	*	*	*	*	*
was	NONE	was	O	0	0	0	1	0	0
truly	NONE	truly	O	0	0	0	1	0	0
aware	NONE	aware	O	0	0	0	1	0	0

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